

SAURAV KUMAR

Cybersecurity Analyst | Application Security | Full-Stack Developer
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Professional Summary

Cybersecurity-focused Full-Stack Developer with hands-on experience in application security, secure software development, vulnerability assessment and penetration testing (VAPT), and AI-powered engineering. Track record of leading security initiatives for 700+ participants, shipping production full-stack applications, and publishing peer-reviewed security research in IEEE Xplore.

Core Skills

Programming Languages: Python, TypeScript, JavaScript, SQL, Bash

Cybersecurity: Application Security, Web Security, Vulnerability Assessment and Penetration Testing (VAPT), Threat Modeling, OWASP Top 10, Capture the Flag (CTF), IoT Security

Development: Full-Stack Development, REST APIs, Next.js, React, Express.js

Cloud & DevOps: Microsoft Azure, Docker, Git, GitHub, Linux, VPS Management, CI/CD

AI & Machine Learning: Large Language Models (LLMs), Computer Vision, Prompt Engineering

Databases: MySQL, MongoDB, PostgreSQL, SQLite

Platforms: Linux, Windows

Professional Experience

DCG Gurugram — Partner, Cybersecurity Community Operations

Aug 2025 – Present

- Lead planning and execution of cybersecurity events, workshops, and technical meetups, partnering with industry professionals to expand community engagement.
- Drive cybersecurity awareness through structured outreach to student and professional communities.
- Focus areas: Community Management, Event Planning, Cybersecurity Awareness.

MRISA (Manav Rachna InfoSec Army) — IoT Security Head

Aug 2024 – Present

- Lead IoT security initiatives and mentor a team of student security researchers.
- Organized Capture the Flag (CTF) competitions for 700+ participants, overseeing logistics and challenge design.
- Authored web exploitation challenges used in hands-on security training labs.
- Technologies: Linux, CTF, Web Security, IoT Security.

Projects

[Defendrix](#) — Advanced Web Application Vulnerability Scanner | Python, Flask, Selenium, REST APIs, MySQL

- Engineered an automated OWASP Top 10 vulnerability scanner that accelerates early-stage security assessment by automating endpoint discovery and attack surface mapping, cutting manual testing effort.
- Architected reusable scanning modules with structured, actionable vulnerability reporting.

[SMEH News Platform](#) — Secure Full-Stack News CMS | Next.js 15, React 19, Express.js, Prisma, SQLite, PostgreSQL, Docker, PM2, Linux VPS

- Built a secure, full-stack news publishing platform that streamlines content management and rapid distribution of institutional news.
- Automated deployment, updates, and server recovery on a Linux VPS, improving reliability and reducing manual maintenance.
- Hardened the platform with JWT authentication, Role-Based Access Control (RBAC), secure file validation, and HTTP security best practices; architected a scalable schema with zero-downtime migration via Prisma ORM.

[AssistAI](#) — AI Accessibility Companion / Next.js, TypeScript, Firebase Genkit, Google Gemini, Tailwind CSS

- Built an AI-powered accessibility platform that improves digital inclusion through conversational AI, real-time vision assistance, sign language recognition, and multimodal object/scene understanding.
- Designed an accessible, keyboard-navigable interface backed by scalable server-side AI inference.

[KnoxGuard](#) — URL Threat Detection System / Python, Flask, Selenium, MySQL

- Built a desktop-based URL threat detection system that flags phishing and malicious websites via rule-based validation, delivering real-time alerts through a lightweight, modular detection engine.

Research Publication

"AI-Driven Techniques for Web Search Vulnerability Identification" — Co-authors: Alan Jolly John, Sarthak Dubey.
Published in IEEE Xplore, 2026. ieeexplore.ieee.org/document/11386307

Education

Bachelor of Computer Applications (BCA)

2023 – Present

Manav Rachna International Institute of Research and Studies (MRIIRS), Faridabad, Haryana, India

Awards & Achievements

- Winner — INTRUSIONX, GLA University (2025)
- 8th Place — CyCog CTF, IEEE MRU & MRU (2024)
- Finalist — Avinya Tech Fest, IIT Guwahati (2024)
- Winner — Hacksplash 1.0, Echelon Institute (2023)
- Ranked 170/1500+ — CyberHavoc CTF (2023)
- 3rd Place — Tech Trover, Debugging Competition (2023)